Lab 7: Fuzzing for Input Validation Bugs

Name: Sai Likhith S

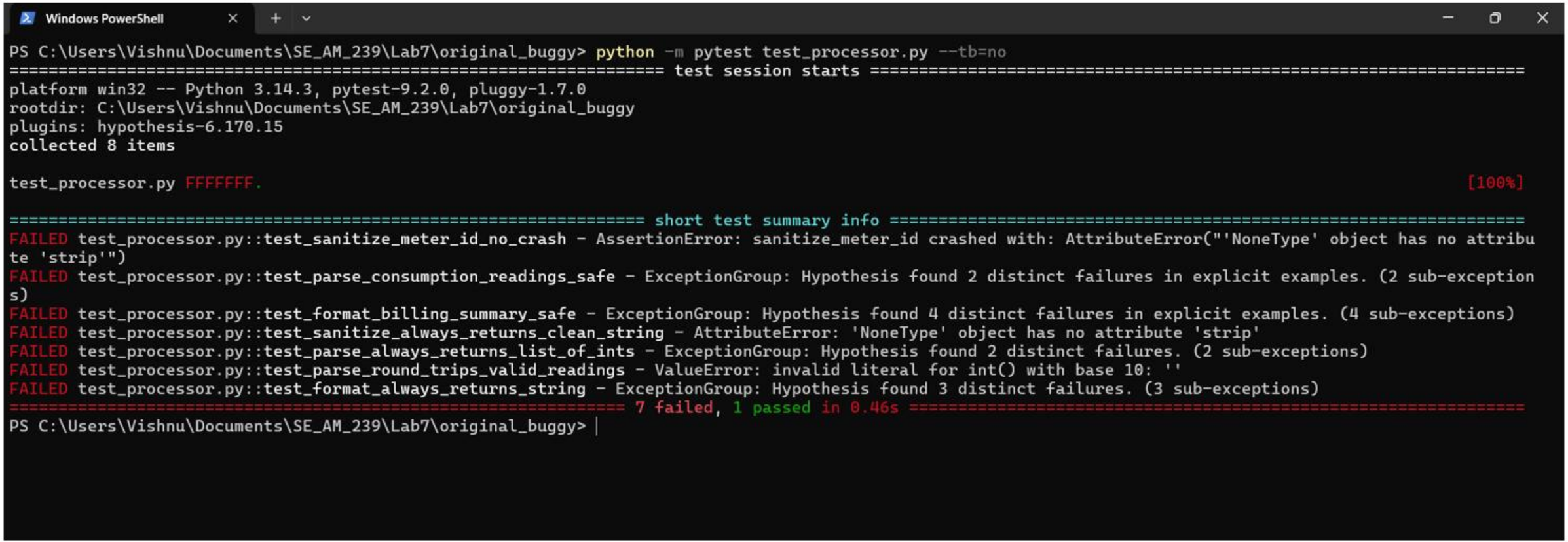
SRN: PES1UG24AM239

Step 2: Buggy processor.py - python processor.py

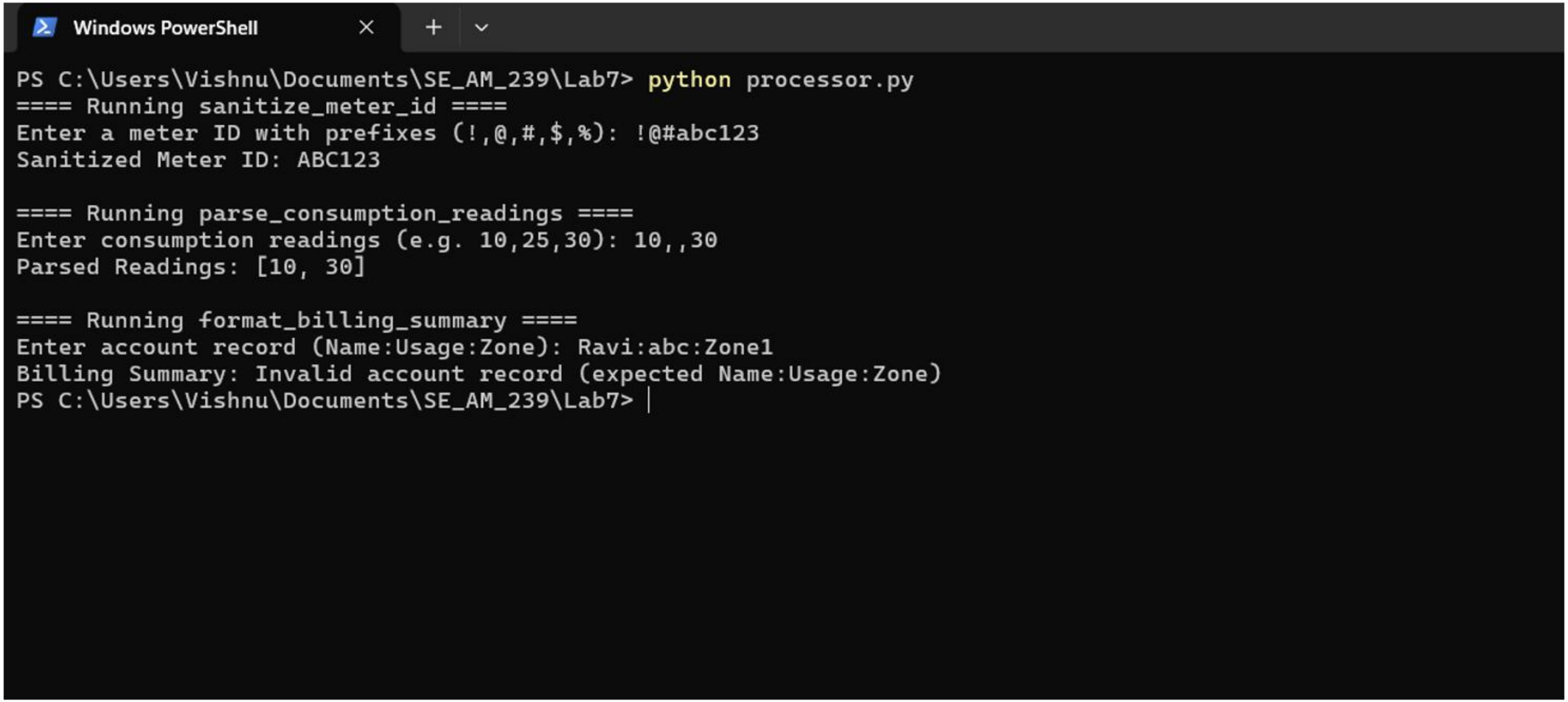
```
PS C:\Users\Vishnu\Documents\SE_AM_239\Lab7\original_buggy> python processor.py
==== Running sanitize_meter_id ====
Enter a meter ID with prefixes (!,@,#,$,%): !@#abc123
Sanitized Meter ID: ABC123

==== Running parse_consumption_readings ====
Enter consumption readings (e.g. 10,25,30): 10,,30
Traceback (most recent call last):
  File "C:\Users\Vishnu\Documents\SE_AM_239\Lab7\original_buggy\processor.py", line 35, in <module>
    main()
    ~~~~^^
  File "C:\Users\Vishnu\Documents\SE_AM_239\Lab7\original_buggy\processor.py", line 26, in main
    readings_list = parse_consumption_readings(csv_input)
  File "C:\Users\Vishnu\Documents\SE_AM_239\Lab7\original_buggy\processor.py", line 9, in parse_consumption_readings
    return [int(r) for r in readings]
            ~~~^^^
ValueError: invalid literal for int() with base 10: ''
PS C:\Users\Vishnu\Documents\SE_AM_239\Lab7\original_buggy>
```

Step 4: pytest test_processor.py against the buggy processor.py (tests failing)



Step 5: Fixed processor.py - python processor.py



Step 6: pytest test_processor.py against the fixed processor.py (tests passing)

```
PS C:\Users\Vishnu\Documents\SE_AM_239\Lab7> python -m pytest test_processor.py --tb=no
============================= test session starts ==============================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Vishnu\Documents\SE_AM_239\Lab7
plugins: hypothesis-6.170.15
collected 8 items

test_processor.py ........                                               [100%]

============================== 8 passed in 0.68s ===============================
PS C:\Users\Vishnu\Documents\SE_AM_239\Lab7>
```